COHOMOLOGY ON SEMI-REPRESENTABLE COMPONENTS IS COMPUTED FACTORWISE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: propagation of lemma-sanity-check (this batch, L46308) through the product decomposition; pairing note recorded.

Editorial note. The label lemma-sanity-check-simplicial-semi-representable names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Base sanity check. Same unit, Lemma lemma-sanity-check. H^p(U, F) = H^p(U, g_n^{-1}F) on a single component, the statement being propagated.
- Product decomposition. Same unit, Section section-semi-representable. Sh(C/K_n) = product of Sh(C/U_{n,i}), which splits the K_n-level statement into factors.

Source reference. Original reference 0DBZ: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1883–1883. Prerequisite pointer invoked at proof line(s) 4037; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 09WK: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3578–3578. Prerequisite pointer invoked at proof line(s) 4032; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let K be a simplicial object of $SR(\mathcal{C})$. Let $U/U_{n,i}$ be an object of $\mathcal{C}/K_n$. Let $\mathcal{F} \in Ab((\mathcal{C}/K)_{total})$. Then*

$$H^p(U, \mathcal{F}) = H^p(U, \mathcal{F}_{n,i})$$

where

(1) *on the left hand side U is viewed as an object of $\mathcal{C}_{total}$, and*

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *on the right hand side* $\mathcal{F}_{n,i}$ *is the* i*th component of the sheaf* $\mathcal{F}_n$ *on* $\mathcal{C}/K_n$ *in the decomposition* $Sh(\mathcal{C}/K_n) = \prod Sh(\mathcal{C}/U_{n,i})$ *of Section 09WK.*

Proof. This follows immediately from Lemma 0DBZ and the product decompositions of Section 09WK. □